AMRITA SCHOOL OF COMPUTING • CHENNAI

QUANTUMSAFE VPN

Post-Quantum Cryptography Secured Android Virtual Private Network

HERE'S HOW TO USE IT:

1. Launch Android App & Configure Gateway IP
2. Authenticate Server with ML-DSA-65
3. Ephemeral KEM Handshake (ML-KEM-1024)
4. One-Tap Encrypted Tunnel & Live Telemetry

PROJECT INVESTIGATORS

P. Abishek Singh (U4CYS23001)

R. S. Sanjaay Krishna (U4CYS23043)

Supervisor: Dr. G. Saranya, Asst. Prof.

CODE BASE

RELEASES

Project Overview

A snapshot of the what, why, and how

QuantumSafe VPN is an end-to-end post-quantum cryptographic VPN engineered to protect mobile IP communications against *Harvest Now, Decrypt Later (HNDL)* attacks and future cryptanalytically relevant quantum computers running Shor's algorithm.

By superseding vulnerable RSA and elliptic curve Diffie-Hellman primitives with finalized NIST standards (**FIPS 203 ML-KEM-1024** & **FIPS 204 ML-DSA-65**), the system delivers authenticated, hardware-accelerated **AES-256-GCM** encrypted tunneling between an unprivileged Android client and a hardened Linux security gateway with automated dynamic IP pooling.

Goals & KPIs

What are we aiming for?

- Immunizes mobile communications against quantum cryptanalysis using NIST FIPS 203 & 204 standards.
- Zero-leak full tunneling with sub-40ms PQC handshake & real-time cryptographic telemetry.

Architecture Overview

A visual mind map illustrating the architectural components

QUANTUMSAFE VPN

SECURITY & PQC

- ML-KEM-1024 (FIPS 203) Key Encapsulation
- ML-DSA-65 (FIPS 204) Gateway Signature
- Hardware AES-256-GCM AEAD Tunneling
- HKDF-SHA256 (RFC 5869) Key Derivation
- 64-bit Sliding-Window Anti-Replay Defense

TECH STACK

- **Client:** Kotlin, Android SDK 34, Coroutines
- **Server:** Python 3.11, Linux TUN, iptables
- **Crypto:** pure-pqc, BouncyCastle, cryptography
- **Testing:** Wireshark, Scapy, pytest, LMK tests
- **OS:** Kali Linux 2024.2 & Android 14

ANDROID CLIENT

- Native `VpnService` Datagram Interception
- Pure-Kotlin Keccak & Lattice Cryptography
- Kernel Socket `protect()` Loop Isolation
- Jetpack MVVM & Material Design 3 UI
- Live Cryptographic Telemetry Dashboard

LINUX GATEWAY

- Linux Kernel Virtual TUN Interface (`tun0`)
- Dynamic IPv4 Lease Subnet (`10.8.0.0/24`)
- Kernel `iptables` NAT Masquerading
- Length-Prefixed TCP Stream on Port 7000
- High-Throughput Non-Blocking Packet I/O

Impact & Conclusion

highlights the value and quantum-resilient defense outcome

QuantumSafe VPN establishes a breakthrough mobile implementation of post-quantum cryptography, proving that lattice-based security standards (**FIPS 203 & 204**) can be deployed on resource-constrained Android smartphones without compromising throughput, battery endurance, or user experience.

By resolving critical platform challenges—such as socket routing loops, Low Memory Killer (LMK) eviction, and Keccak CPU bottlenecks—this project delivers a practical blueprint for mobile zero-trust enterprise security, safeguarding vital data against state-sponsored quantum decryption.

- ✓ Future-Proof Quantum Immunity
- ✓ Zero Architectural Leaks
- ✓ Battery-Optimized Multithreading

Insights & Results

what the platform verifies, defends, and delivers

Testing on a physical **Google Pixel (Android 14)** connected to a **Kali Linux** gateway verified end-to-end mathematical correctness across all NIST Known-Answer Test (KAT) vectors with a 100% automated test suite pass rate.

Adversary simulation confirmed absolute cryptographic integrity: **100% of ciphertext bit-flipping attacks** were instantly dropped by Galois MAC tags, packet replays were thwarted by sliding-window sequence tracking, and rogue server substitutions were blocked via ML-DSA fingerprint pinning.

100%	100%	0%	~38 ms
KAT COMPLIANCE	TAMPER DEFENSE	DNS / IP LEAKS	PQC HANDSHAKE
FIPS 203 & 204	Bit-Flip Rejection	Full Tunneling	Sub-40ms Ephemeral